

Collection of Exercises for PLC Programming

PREFACE

I am happy to have the opportunity to publish a PLC programming book for the third time. This time, it's a collection of exercises for learning PLC programming.

For the past seven years I have been a trainer for PLC control in the Automation Technology program (a two-year higher education program) at Dania Academy in Denmark; I have developed many exercises for the students, and the best of these are now found in this book.

Basis of knowledge

The ideas for the exercises have come to me by visiting a wide range of trade fairs, updates on social media, literature on automation, as well as from the students' exam projects. Additionally, I get inspiration from my professional work with programming.

Selection of exercises

I have chosen exercises for the book that I believe provide good learning within PLC, and at the same time, exercises that the students find exciting. I have spent a long time developing them to ensure that they cover many of the professional challenges that can arise in PLC programming, as well as ensuring that they cover many professional aspects within PLC programming.

The book is pedagogically and didactically structured with more than 150 illustrations and tables, which means that there is a progression in learning throughout the book, and therefore, the exercises increase in difficulty.

Unfortunately, I am unable to assist with the exercises or provide solutions to them, but comments, critiques, and suggestions for improvements are warmly welcomed via email at: TomMejerAntonsen@gmail.com

I have previously published the following books:

"PLC Control with Structured Text (ST)", first edition 2018

"PLC Control with Ladder Diagram (LD)", first edition 2021

These books are now used in a wide range of educational programs and schools, and many companies have purchased them for their employees. Naturally, tips for some of the exercises can be found in the books.

Thanks to students for review, feedback, and inspiration.

I hope you enjoy the book.

Best regards,

Tom Mejer Antonsen

Randers, Denmark, April 2024

Tom Mejer Antonsen

Collection of Exercises for PLC Programming

100 programming exercises from beginner to expert level

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1 Introduction

This book contains exercises for programmable logic controllers (PLC).

It is pedagogically structured, starting with simple programming exercises involving only logic (relay logic with on/off signals). After this, more programming techniques are added with increasing the difficulty. This means that after the simple logic exercises, exercises with counters, timers, analog sensors, applied mathematics, ARRAY, STRUCT and STRING are included. The book also includes exercises where a safety PLC must be programmed, as well exercises with component and sensor naming, programming of functions and function blocks, sequence technique, manual or automatic operation, integration with a robot, frequency converters (VFD), or vision cameras.

The final chapter, chapter 11, consists of mixed complex exercises where all programming techniques are combined. For example, there are exercises for production lines and data collection, as well as ones in which sensors must be selected, and the control specification must be written before programming.

All exercises in the book are selected to be relevant, practical, and realistic in relation to the typical challenges that may arise with PLC programming.

Getting started with the exercises

If you are a beginner in PLC programming, it is recommended to start with the first exercises in the book. The initial exercises are at a beginner level and provide a good introduction to programming in a PLC. You can go to the more complex exercises later in the book as you gain more experience.

Choice of PLC brand and programming language

The exercises can be solved in all types of PLCs, regardless of brand or model. Similarly, there is no requirement to use a specific PLC programming language. You will gain a lot of learning by solving the tasks in several differing PLC programming languages. For example, an exercise can be programmed first in Ladder Diagram (LD) and then in Structured Text (ST).

Some exercises will naturally be more suitable for a specific programming language, so it is up to you to choose the right one. Similarly, some exercises are best for using sequence techniques and developing own functions.

Who can use this book?

The book is primarily designed for use for the higher education Academy Profession (AP) graduate in Automation Engineering. However, the book can also be used for many other educations that include PLC programming. This could be electrician training, control and regulation electrician training and automation technicians, as well as higher education programs for mechatronics engineers or automation engineers.

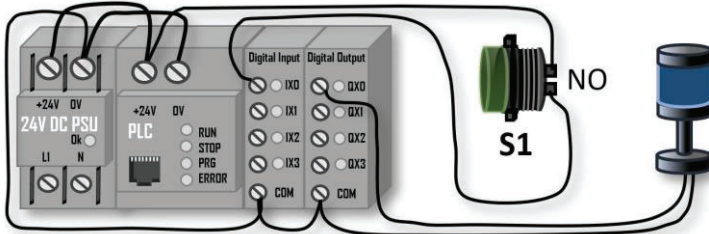
Sorry, no guarantee, support, or solutions are provided for the exercises in the book.

2 Logical circuit exercises

This chapter contains exercises with logical circuits only; the first exercises are at beginner-level. Later in the chapter, more challenging exercises will be introduced. All tasks can be solved using relay logic (on/off signals).

2.1 Turn on light in lamp with a push button

This exercise has a push button switch **S1** and a lamp **P1** connected to a PLC:



The push button switch **S1** is connected to a digital input module, and the lamp **P1** is connected to a digital output module.

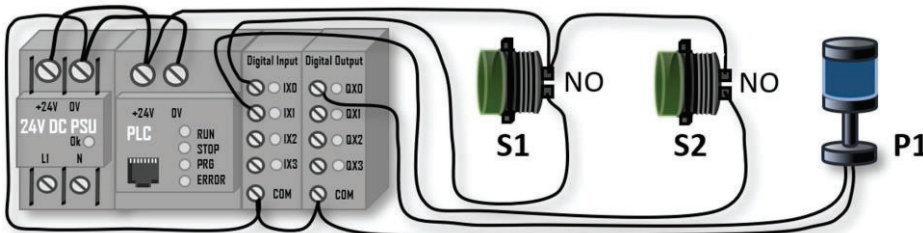
The push button switch is a Normally Open (NO) contact.

Write a PLC program that meets the following requirements:

- The lamp **P1** is turned on when **S1** is pressed.
- The lamp **P1** should be off when **S1** is not pressed.

2.2 Turn on light in lamp with two push buttons

Here are two push buttons and a lamp connected to a PLC:



Both push buttons are Normally Open (NO) contacts.

The two push buttons are connected to separate digital input on a digital input module, and the lamp is connected to a digital output module.

Write a PLC program where lamp **P1** is turned on when both push buttons **S1** and **S2** are pressed simultaneously (at the same time). If both push buttons are not pressed simultaneously, the lamp should remain off.